Practice sheet: 1

DSA lab

```
 // q1. Largest element in array
import java.util.*;
public class max {
public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);
      int n = sc.nextInt();
      int[] arr = new int[n];
      for(int i = 0; i < n; i++) {
          arr[i] = sc.nextInt();
      }
      int max = arr[0];
      for(int i = 1; i < n; i++) {
          if(arr[i] > max) {
              max = arr[i];
          }
      }
      System.out.println("Largest Element = " + max);
}
}
```

Problem 2: Second Largest Element in an Array

```java
import java.util.Scanner;
public class Second {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        int[] arr = new int[n];
        for(int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
        }
        int largest = Integer.MIN_VALUE;
        int secondLargest = Integer.MIN_VALUE;
        for(int i = 0; i < n; i++) {
            if(arr[i] > largest) {
                secondLargest = largest;
                largest = arr[i];
            }
            else if(arr[i] > secondLargest && arr[i] != largest) {
                secondLargest = arr[i];
            }
        }

        System.out.println("Second Largest = " + secondLargest);
    }
}
```

Problem 3: Check if Array is Sorted

```java
import java.util.Scanner;
```

```java
public class Sort {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();
        int[] arr = new int[n];

        for(int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
        }

        boolean sorted = true;

        for(int i = 0; i < n - 1; i++) {
            if(arr[i] > arr[i + 1]) {
                sorted = false;
                break;
            }
        }

        System.out.println(sorted);
    }
}
```

Problem 4: Reverse an Array

```java
import java.util.Scanner;
```

```java
public class Reverse {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();
        int[] arr = new int[n];

        for(int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
        }
        int left = 0;
        int right = n - 1;
        while(left < right) {
            int temp = arr[left];
            arr[left] = arr[right];
            arr[right] = temp;
            left++;
            right--;
        }
        for(int num : arr) {
            System.out.print(num + " ");
        }
    }
}
```

Problem 5: Linear Search

```java
import java.util.Scanner;
```

```
public class Search {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();
        int[] arr = new int[n];

        for(int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
        }

        int key = sc.nextInt();
        int index = -1;
        for(int i = 0; i < n; i++) {
            if(arr[i] == key) {
                index = i;
                break;
            }
        }
        System.out.println(index);
    }
}
```

Problem 6: Left Rotate Array by One Place

```
import java.util.Scanner;

public class Left {
```

```
    public static void main(String[] args) {

        Scanner  sc = new Scanner(System.in);
        int n = sc.nextInt();
        int[] arr = new int[n];

        for(int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
        }
        int first = arr[0];
        for(int i = 0; i < n - 1; i++) {
            arr[i] = arr[i + 1];
        }
        arr[n - 1] = first;
        for(int num : arr) {
            System.out.print(num + " ");
        }
    }
}
```

Problem 7: Move Zeroes to End

```
import java.util.Scanner;

public class Zeroes {
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        int[] arr = new int[n];
        for(int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
        }
        int j = 0;
        for(int i = 0; i < n; i++) {
            if(arr[i] != 0) {

                int temp = arr[i];
                arr[i] = arr[j];
                arr[j] = temp;
                j++;
            }
        }
        for(int num : arr) {
            System.out.print(num + " ");
        }
    }
}
```

Problem 8: Remove Duplicates from Sorted

```
import java.util.Scanner;

public class RemoveDuplicates {
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();
        int[] arr = new int[n];

        for(int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
        }

        int k = 0;
        for(int i = 1; i < n; i++) {
            if(arr[i] != arr[k]) {
                k++;
                arr[k] = arr[i];
            }
        }
        for(int i = 0; i <= k; i++) {
            System.out.print(arr[i] + " ");
        }
        System.out.println("\nCount = " + (k + 1));
    }
}
```

Problem 10: Missing Number

```
import java.util.Scanner;

public class Miss{
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);
```

```
        int n = sc.nextInt();

        int[] arr = new int[n];

        int sum = 0;

        for(int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
            sum += arr[i];
        }

        int expected = (n * (n + 1)) / 2;

        System.out.println("Missing Number = " + (expected - sum));
    }
}
```

End of sheet 1

Thank you